IoT based ECG System for Arrhythmia Detection using Telemetry Monitoring

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Abstract—An electrocardiogram (ECG) is the primary signal used for the detection of any cardiovascular abnormalities, but conventional clinical ECG scans include restrained time-duration ECG signals, lacking the asymptomatic or essential attributes of CVDs. Therefore, developing a system for real-time and long-term ECG monitoring is crucial for the early diagnosis of CVDs. With technological advancements in the field of wearable electronics and IoT innovations, developing a system for convenient and long-term ECG monitoring is easily possible. In this paper, we suggest a new approach for ECG measurement centered on Internet of Things (IoT) schemes that can facilitate long-term monitoring. The required ECG information is acquired utilizing a wearable observing node consisting of an AD8232 ECG sensor with AgCl gel based electrodes. The conditioned ECG signal is transmitted to the IoT cloud utilizing Wi-Fi technology. We have used the Node MCU ESP8266 Wi-Fi module to transmit the ECG data to the IoT cloud and the Blynk 2.0 as an IoT platform to display that data. The program code was developed to acquire ECG waves along with important features like R-R interval, QRS duration, PR interval, QT interval, HRV, and heart rate. These important diagnostic parameters are displayed on the IoT channel with the required ECG information for the patient and medical professionals. It can be used by medical experts anywhere across the globe to analyze and diagnose the cardiac health conditions of the subjects under investigation and may provide necessary emergency health assistance. The obtained results on the Blynk IoT channel are compared with the clinical ECG of the same subject; the results are acceptable with accuracy of more than 95%, and in compliance with the standard ECG. It can be used for the detection of arrhythmia conditions. We believe the developed system is a positive contribution to telemedicine.

Keywords— Internet of Things (IoT), AD 8232 ECG Sensor, Node MCU, Blynk 2.0, ECG signals, Heart rate, Heart rate variability (HRV).

I. Introduction

Cardiovascular diseases are one of the prevalent and dangerous ailments that influence human health. The detection of cardiac disease in its early stages by the use of early detection is a major problem in today's scenario [18]. Cardiovascular diseases (CVDs) are the leading cause of death globally (32.1%) and result in 17.9 million deaths in 2015 [1,13]. However, it is estimated that 90% of CVDs is preventable, The sooner disease is identified, the better preventive and curative care can be ensured [1,14]. Therefore a convenient and cost-effective system for long-term ECG monitoring is highly desirable.

Healthcare has become one of the major concerns for the people and governments due to the increasing rise of the human population and medical expenditure. Meanwhile, the problem of ageing populations is becoming more serious, according to a study by the World Health Organization (WHO). Elderly people's health needs to be monitored more regularly in general, offering a greater challenge to current medical systems. As a result, there has been a growing focus on identifying human diseases rapidly and effectively at a minimal cost. In hospitals and medical research, ECG monitoring is extensively used due to its dominance in the detection of cardiac diseases. Conventionally, ECGs have been performed in professional medical facilities using huge stationary equipment. For their good short term measurement accuracy, these types of equipment commonly use 12 lead electrodes to capture ECG data. However, because the equipment is not likely to be portable, thus subject activities during ECG collection phase are severely limited. Furthermore, because these systems are very costly to use at home patients will be forced to attend the hospitals and clinics more regularly, increasing the hospital's workload.

The developments in wireless sensor networks (WSN), body area networks (BAN), cloud computing and big data technologies trigger the use of Internet of Things (IoT) in healthcare industry [17]. Portable IoT enabled ECG monitoring systems have emerged as a result of the development of wireless sensor network (WSN) and mobile internet [2,15], which can sense ECG signals using a non-contact sensor and transmit the signal to the IoT platform using wireless communication protocols such as Wi-Fi, Bluetooth and Zigbee [15]. WSN-based ECG monitoring devices use lesser electrodes than typical ECG monitoring systems for mobility a reason which leads to inaccuracy. At the expense of accuracy it is sufficient to obtain the necessary of the heart. These portable systems are typically embedded in some wearable materials and have a minimal effect on the user's daily routines. The real-time and long-term ECG can be easily monitored with the use of these systems.

This work focuses on developing a low-cost ECG monitoring system using IoT techniques. The system is composed of a single-lead AD 8232 ECG sensor which acquires the ECG signal from the human body and sends the ECG data in analog form serially to the Node MCU microcontroller, which then converts the analog data into digital form using an inbuilt ADC. The digital data is then processed and several key features of ECG wave like R-R Interval, QRS duration, PR interval and QT interval are

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extracted from the measured ECG signal. After extracting these key features the program will do the calculations of heart rate and heart rate variability (HRV) along with this the program will also inspect the ECG data for detecting arrhythmia conditions such as tachycardia where the heart rate in bpm is greater than 100 bpm and bradycardia where heart rate is less than 60bpm. The collected information is then sent legitimately to the Blynk 2.0 IoT cloud using on-chip Wi-Fi in Node MCU microcontroller. Blynk mobile app and Blynk Web dashboard are used as GUI to visualize the collected ECG data

II. Architecture Of The Proposed IoT Based ECG System

In this section, we defined the architecture of the proposed system which provides IoT driven long term and continuous ECG monitoring for the early detection of cardiovascular diseases. The IoT based ECG system consists of three typical IoT components [2, 16], i.e., the measurement node, the IoT cloud and the Graphical user interface (GUI). Figure 1 shows the Architecture of the developed IoT based ECG System.

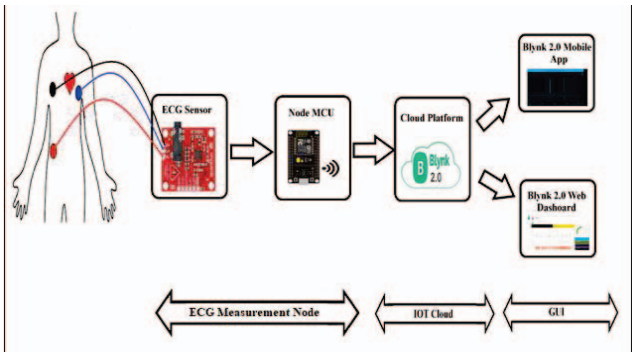

Fig. 1. Architecture of the developed IoT based ECG System.

A. ECG Measurement Node

The ECG measurement node is the module responsible for acquiring ECG signal data from the subject under investigation and then sending it to the IoT cloud via a wireless channel such as Wi-Fi. The ECG measurement node mainly consists of an ECG sensor to acquire ECG signal, a microcontroller to process the measured ECG signal obtained from the sensor and a Wi-Fi module to send the data to the IoT cloud. We used AD 8232 single lead ECG sensor for its very low power requirement and very high noise rejection capability, low power requirement is very necessary for the application of long term continuous ECG monitoring. Node MCU is a reliable and cost-effective for microcontroller IoT applications [3]. The processing capability of Node MCU is enough to perform the calculations of key features and analysis of ECG data for arrhythmia detection. USB and lithium-ion batteries are the two possible ways for powering the ECG sensor and Node MCU microcontroller. A lithium-ion battery is preferred because of its extended life and portability.

B. IoT Cloud

As soon as the system receives bulk of ECG data from the monitoring node, an IoT cloud is needed to save this data to a database quickly and cost-effectively and display the measured results as needed. An IoT cloud is created using cloud computing, data storage and web services. We used Blynk 2.0 IoT cloud for its very simple setup and interactive GUI [4]. Blynk runs on all platforms, including Android, iOS, and Web, and can operate a variety of hardware modules, including Arduino, Node MCU, Raspberry Pi, and over 500 others. WiFi, Ethernet, USB, serial, and Bluetooth are among the communication options available to link the hardware module device to the Internet. Blynk allows you to create a large number of applications also known as templates in Blynk and use it to control and monitor several devices simultaneously provided with internet access from anywhere and anytime around the globe. The figure 2 illustrates Blynk 2.0 Cloud Architecture.

Fig. 2. Blynk 2.0 Cloud Architecture.

C. Graphical User Interface (GUI)

The graphical interface is responsible for display and management of the measured ECG data. It facilitates easy access to data to the concerned users. Users can connect to the cloud to capture the displayed ECG data in real-time. Preferably, there are two types of graphical interfaces that allow users to view ECG data, namely mobile web pages and mobile apps. The Blynk IoT platform provides two GUI options one is the Blynk Mobile Application available for both Android and IOS users and the other is a web-based dashboard to visualize and analyze the data obtained from the IoT devices. The GUI provides a user-friendly dashboard where we can choose different widgets like charts, labels, clocks, buttons etc.

III. Implementation Of The Proposed System

The IoT based ECG system is implemented according to the proposed architecture in the section "Architecture of the proposed IoT based ECG System". The hardware of the system as shown in fig. 3 is a very simple circuit consisting of AD8232 ECG Sensor and Node MCU microcontroller which is implemented on a simple breadboard and can be incorporated in any day to day usage things like a belt, t-shirts, wrist bands, car steering, office chair etc which makes the system portable and is ideal for a system which is to be used for continuous long term ECG monitoring. The ECG signal is acquired from the human body using the conventional Ag Cl gel electrodes with AD8232 ECG sensor and then the analog data is sent serially to the Node MCU microcontroller which receives it on its analog pin A0 and using a 10 bit inbuilt ADC converts the analog data into digital form for further processing. The digital data is then processed according to a program uploaded in the Node MCU microcontroller, the

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processed ECG data is then sent to the Blynk IoT cloud using Wi-Fi.

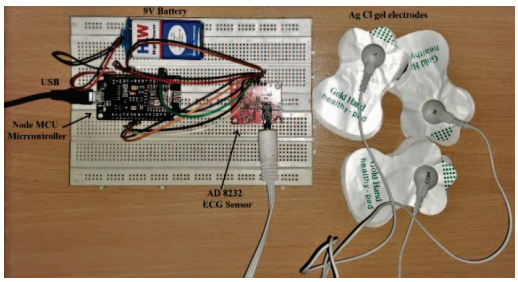

Fig. 3. Circuit of the proposed IoT based ECG system.

Several features of ECG wave like heart rate, R-R interval, QRS duration, PR interval, QT interval and Heart rate variability (HRV) are calculated using different algorithms written in the uploaded program in the microcontroller.

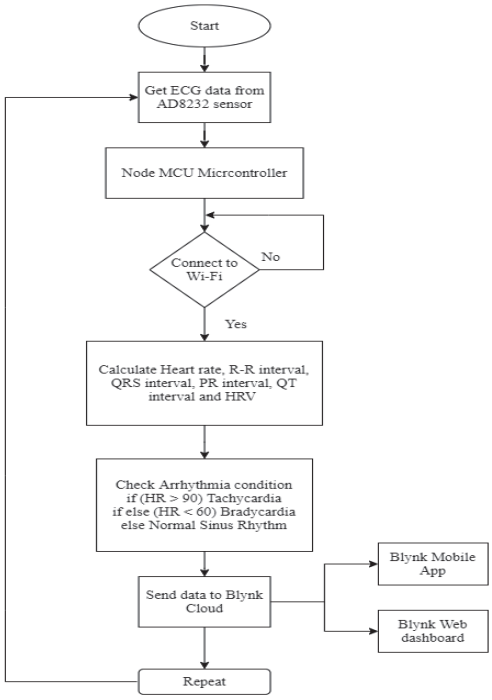

Fig. 4. Flowchart of the Implemented IoT based ECG System.

As depicted from the flowchart of the implemented system in Fig. 4 at first the ECG signal is measured from the human body using the single lead AD 8232 ECG sensor, the sensed signal is then sent to the Node MCU microcontroller which receives it in the form of analog input and then converts it to digital form using an inbuilt ADC. The digital data is then processed according to predefined algorithms written in the program uploaded to the Node MCU microcontroller. The program checks the connection status of the system with Wi-Fi if it finds the system connected, the calculations of key features of the ECG wave along with the heart rate in beats per minute (bpm) and heart rate variability (HRV) will be carried on and finally, the program will analyze the results to check the heart rhythm and to detect arrhythmia conditions such as tachycardia and bradycardia [9].

The obtained results from the calculations along with the measured ECG signal is sent to the IoT cloud where it is visualized in the web-based GUI. The web-based GUI setup is done by following the simple steps as shown in Fig 5, first, we need to sign up for a new account on blynk.cloud website then we need to create a Blynk template in which we have to choose the hardware as Node MCU and Wi-Fi in the type of connection, after this, the Blynk will generate a unique 12 digit alphanumeric template ID and a device name that needs to be included in the program uploaded to the Node MCU microcontroller which tells the IoT device where to publish the data. For the visualization of ECG signal and its key features along with the arrhythmia detection we need to create several data streams in the Blynk template, the data streams are the variables that we need to include in the program at the controller side. Finally by selecting the widgets available in the Blynk template such as charts and labels the data can be visualized.

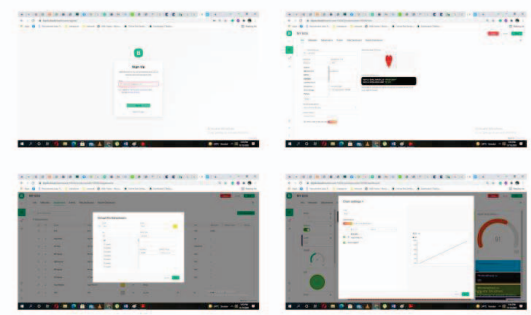

Fig. 5. Blynk Web GUI setup for data visualization.

IV. Results And Discussion

The performance of the proposed ECG system was first analyzed on the Digital Storage Oscilloscope (DSO) and serial monitor of Arduino IDE as shown in fig. 5 to test the working of the ECG sensor and to ensure the proper placement of electrodes. In a relaxed sitting position, we plotted the response obtained from the ECG sensor. From the obtained ECG waveform on the serial plotter of Arduino IDE, we did manual calculations of ECG parameters like R-R interval, QRS duration, PR interval, and QT interval then we developed some algorithm based on these calculations to automate the process of finding these ECG parameters using the program we uploaded in the Node MCU microcontroller.

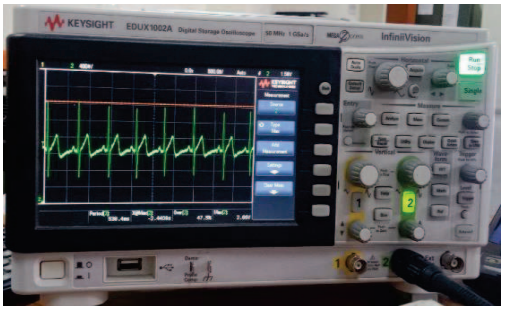

Fig. 6. Response of the system on DSO.

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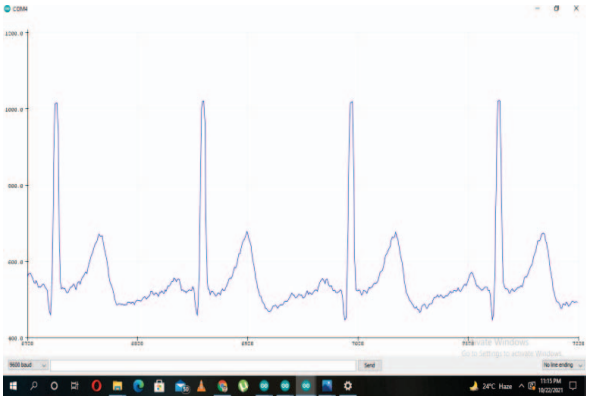

Fig. 7. Response of the system on serial plotter of Arduino IDE.

For the manual calculation of ECG parameters, we obtained the ECG signal data points from the serial monitor of Arduino IDE along with their time stamp and the data is copied to the excel and a graph is plotted between the measured ECG data points in volts on the y-axis and time on the x-axis as shown in fig. 6.

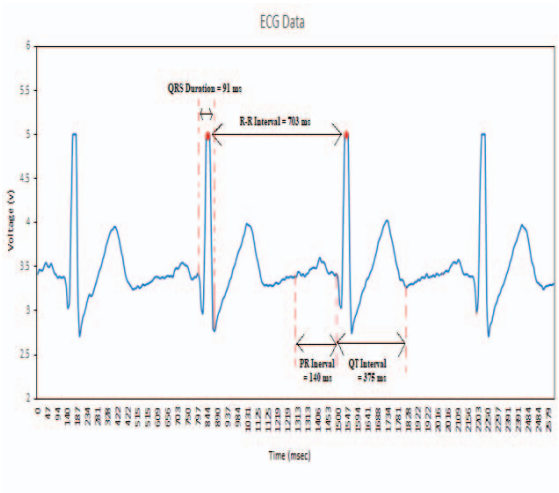

Fig. 8. Key features of the measured ECG signal.

Once we calculated the key features of ECG wave manually we developed some algorithms for the calculations of these features by the microcontroller. The R-R interval is calculated using the algorithm which is based on initializing the inbuilt timer in the microcontroller once the ECG signal crosses the upper threshold which is set for R peak and calculate the time interval between the two successive R peaks which is nothing but the R-R interval. Similarly, PR interval, QT interval and QRS duration are calculated. The calculated R-R interval is then used to find the heart rate in beats per minute which is given as the bpm = (60/R-R interval in seconds) and heart rate variability (HRV) which is given as the difference between the two successive R-R intervals. Once the program calculates the heart rate in bpm a simple algorithm is implemented to detect the arrhythmia conditions such as tachycardia and bradycardia according to the normal threshold values of the heart rate which is between 60 bpm to 100 bpm for normal sinus rhythm, greater than 100 bpm for tachycardia and bradycardia is less than 60bpm.

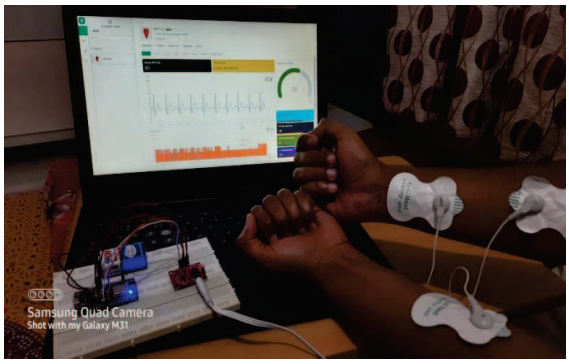

Fig. 9. Demonstration of the proposed IoT based ECG System.

After the successful implementation of all the algorithms and processing the ECG data is finally sent to the Blynk cloud, from where it is visualized in the Blynk web GUI dashboard and the mobile application as shown in fig.8 and fig.9. The obtained results on the GUI are thus compared with the 12 lead clinical ECG and the %ge error is calculated for all the parameters of ECG wave from the proposed IoT based ECG system and the 12 lead clinical ECG as given in table 1.

Fig. 10. ECG data obtained on Blynk Web GUI.

TABLE I. RESULTS COMPARISON SUMMARY OF 12 LEAD ECG AND THE PROPOSED IOT BASED ECG SYSTEM

Parameters	12 lead clinical ECG	Proposed IoT based ECG System	%ge Error
Heart Rate	82 bpm	84 bpm	±2.43
R-R Interval	731 ms	714 ms	±2.32
QRS duration	93 ms	91 ms	±2.15
PR Interval	128 ms	140 ms	±9.37
QT interval	361 ms	375 ms	±3.87
Heart Rhythm	Normal Sinus Rhythm	Normal Sinus Rhythm	

As seen from Table I, it is evident that the average %ge error calculated for all the parameters is found to be ±4.02 % which implies that the proposed system is reliable and in compliance with the 12 lead clinical ECG. The arrhythmia conditions are detected successfully.

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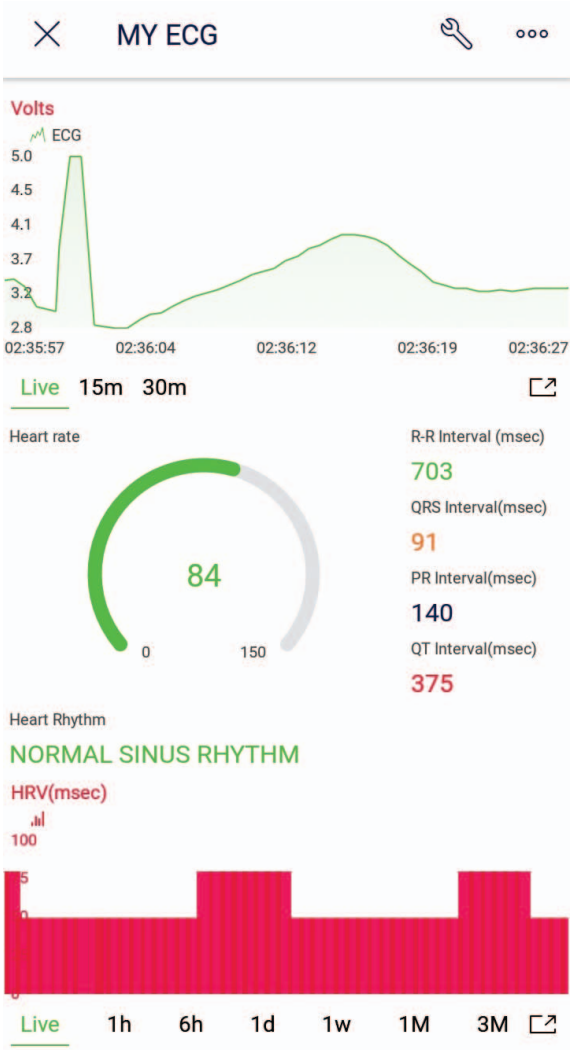

Fig. 11. ECG data obtained on Blynk mobile app.

V. Conclusions

In this work, we developed a low-cost ECG system based on advanced IoT techniques for continuous and long term monitoring of ECG. In this work we present the architecture of the IoT assisted ECG system. The system results are at par with the conventional ECG systems. The ECG signal acquired from the human body using the ECG measurement node consists of AD 8232 ECG sensor with AgCl gel based electrodes and Node MCU microcontroller with on-chip Wi-Fi. The data was sent to the IoT cloud which can store it for long hours and covers a very wide range. Algorithms are developed to calculate the key features of the measured ECG signal and to detect for the arrhythmia conditions. With an interactive web-based and mobile application GUI, the ECG signal and the calculated parameters are visualized along with the status of heart rhythm to detect the arrhythmia conditions. The advantages of the proposed system over the existing systems for ECG measurement are its low cost, high portability and continuous long-term ECG monitoring with arrhythmia detection.

VI. Future Scope

In our future work, we plan to improve the accuracy of the proposed system by embedding the system with multiple leads and capacitive electrodes in a smart vest system. There is room for improvement in reduction of noise and motion artefacts. Improvements in the system could be done for its clinical acceptance.

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